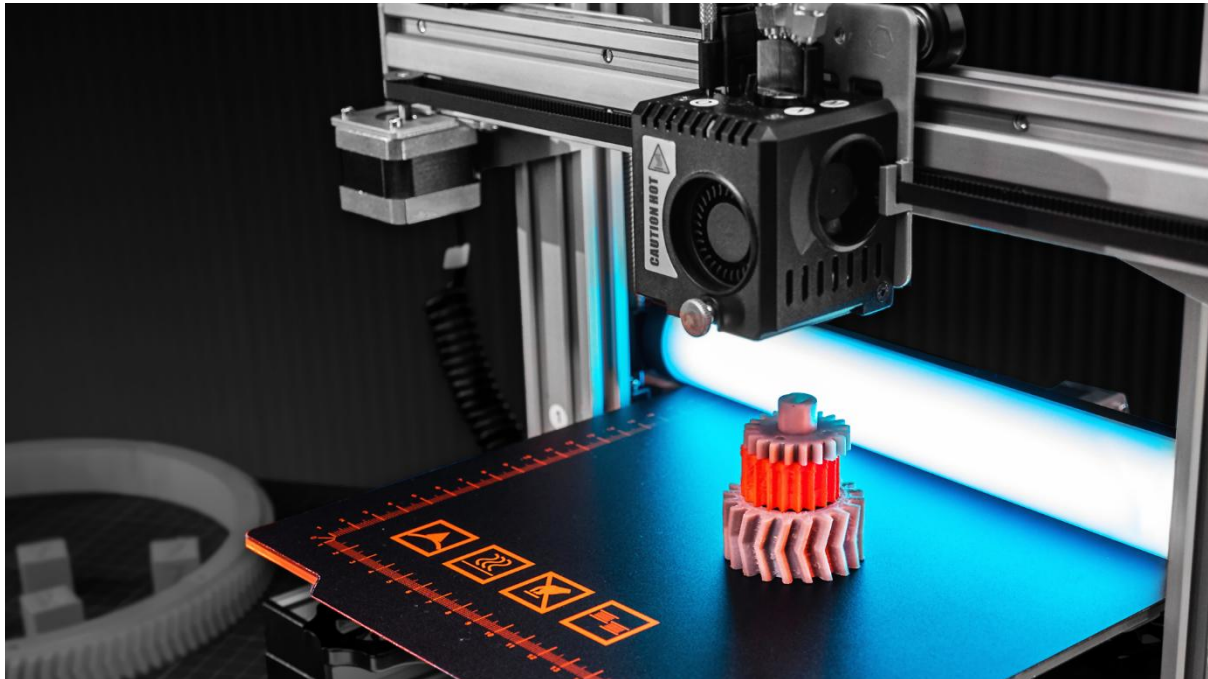




COURSE CONTENT

- 2.1.1 Introduction**
- 2.1.2 Fused Deposition Modelling (FDM)**
- 1.3.3 Applications of FDM**
- 1.3.4 FDM Materials**
- 1.3.5 Process Parameters**

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1.3.1 INTRODUCTION

Solid-based processes use material in solid form, such as filaments (wires), sheets, laminates, or pellets, as the primary feedstock. The fabrication of parts is achieved either by melting and extruding the material or by cutting, stacking, and bonding sheets into successive layers that follow the digital model's geometry. These methods rely on thermomechanical transformation, adhesive bonding, or lamination techniques to create functional parts.

Compared to liquid-based photopolymerization processes, solid-based processes are particularly well-suited for thermoplastics, composites, and laminated structures, offering versatility in both rapid prototyping and functional end-use components. Their advantages include material availability, cost-effectiveness, and suitability for large-scale builds.

Solid-based processes can be categorized into two main groups:

1. Extrusion-Based Processes: In extrusion-based methods, a thermoplastic or composite material in filament, wire, or pellet form is heated, melted, and selectively deposited through a nozzle to build the part layer by layer. The following processes build parts using extrusion:
 - Fused Deposition Modeling (FDM) – Uses thermoplastic filaments extruded through a heated nozzle to create parts.
 - Multi-Jet Modeling (MJM) – Jets multiple streams of molten material often thermoplastic or wax to build detailed parts layer by layer.
2. Lamination-Based Processes: In lamination-based methods, sheets or laminates of material (such as paper, plastic, or composites) are cut to the required shape and bonded together to form the part.
 - Laminated Object Manufacturing (LOM) – Involves cutting and bonding layers of sheet material (paper or plastic foil) with heat and adhesive.
 - Paper Lamination Technology (PLT) – Uses layers of standard or coated paper bonded with adhesive and cut with a laser to form 3D objects.
 - Selective Lamination Composite Object Manufacturing (SLCOM) – Employs composite sheets with fiber reinforcement, laminated and cut to produce high-strength functional parts.
 - Composite Based Additive Manufacturing (CBAM) – Uses sheets of dry fiber reinforcement (carbon or fiberglass), bonded with a polymer matrix and cut to shape, creating lightweight and strong composite structures.

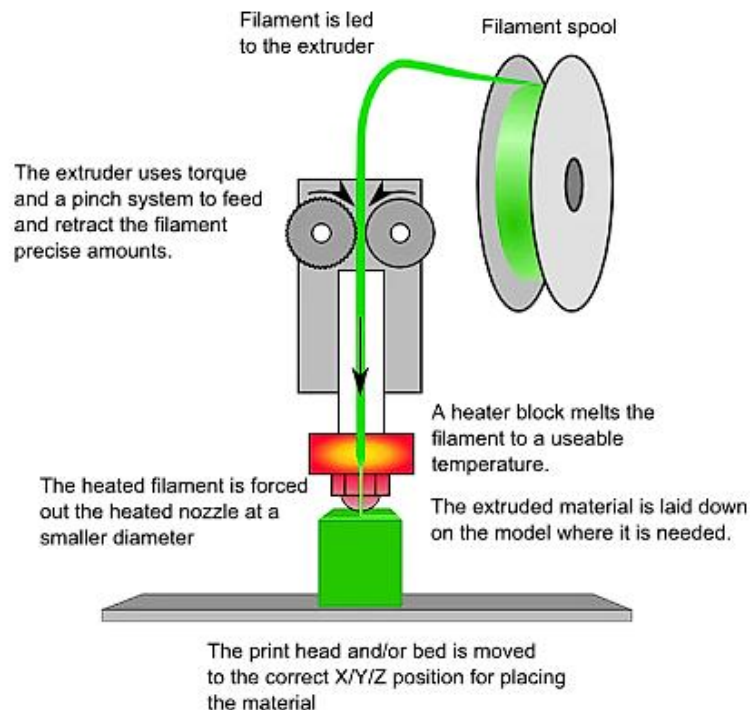
In the present chapter, we will focus specifically on the FDM process, as it is the most widely adopted extrusion-based solid AM technique

1.3.2 FUSED DEPOSITION MODELLING (FDM)

Stratasys Inc., founded in 1989, has played a pioneering role in developing products based on FDM technology. The process was first invented by Scott Crump in 1988, with the patent officially granted in the United States in 1992.

FDM is an extrusion-based rapid prototyping process, although it works on the same layer-by-layer principle as other RP systems. Also known as Fused Filament Fabrication

(FFF). It is one of the most widely used additive manufacturing processes due to its simplicity, cost-effectiveness, and ability to fabricate functional thermoplastic parts.

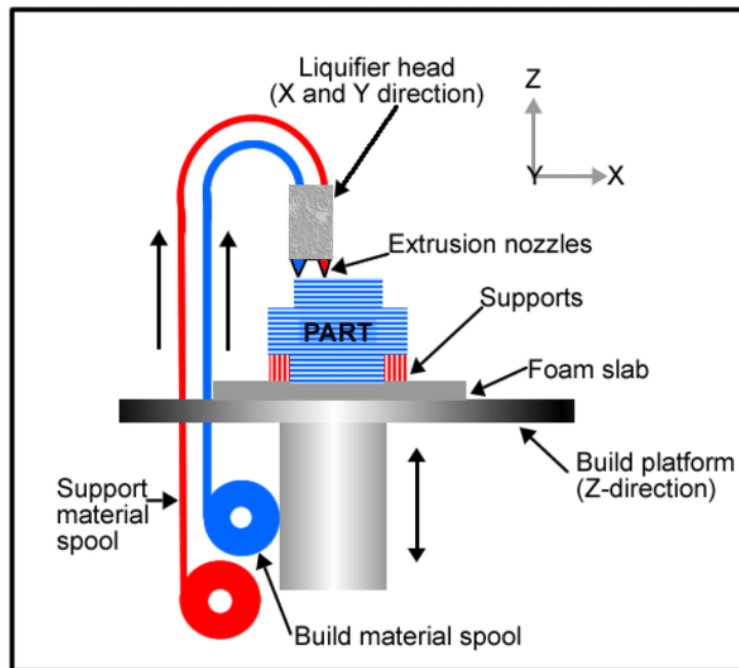


Extrusion principle in FDM

The principle of FDM is based on surface chemistry, thermal energy, and layer manufacturing technology. In this process, a thermoplastic material in the form of a continuous filament spool is fed into a specially designed extrusion head. Inside the head, the filament is heated until it softens or melts, and then it is extruded through a fine nozzle. As the molten material is deposited, it undergoes cooling and solidification, bonding with the previously deposited layer. By repeating this layer-by-layer deposition and solidification cycle, the complete three-dimensional part is fabricated, similar to other AM systems.

HARDWARE

- **Build Platform**: It is the platform on which the part is built layer by layer. It moves in – Z direction.
- **Material spools**: The modeling material is in spools - very much like a fishing line. Both build material and support material are supplied in spool form. The filament on the spools is fed into an extrusion head.
- **Extrusion head**: It heats the material to a semi-liquid state. Two parallel wheels attached to a small electric motor by gears grip the material and feed it to the heaters. The heating element is electronically controlled and has feedback thermocouples to allow for a stable temperature throughout.
- **Extrusion nozzles**: Two Extrusion nozzles (or) tips are externally threaded and screw up into the heating chamber exit. The tips can be removed and replaced with different size openings.



Hardware and Principle of FDM

PROCESS

A geometric model of the part is created using CAD software and saved in STL format. The STL file is sliced into horizontal layers, taking into account part orientation and any necessary support structures. Slice thickness can be adjusted between 0.10 mm to 0.50 mm depending on desired resolution and surface finish. The filament material from the spools is fed into the extrusion head and heated to a semi-liquid state. Dual extrusion systems allow one nozzle to print the primary build material and the other to deposit support material. The molten filament is extruded through the nozzle and deposited as ultrathin layers onto the build platform. The surrounding air is kept below the filament's melting point, allowing the material to solidify rapidly after deposition. The part is built layer by layer, from the bottom up, with each new layer bonding to the previous one through thermal fusion.

ADVANTAGES:

1. **Variety of Material Choice:** Material used for printing is available in various options. Some common of them are PLA, PETG, ABS, PET, TPU and Nylon.
2. **Compact Design:** FDM technology-based printers are quite simple and thus do not take up too much space. The size of the printer is totally dependent on the build volume. And other parts like the outer frame, power unit, etc. don't take much space.
3. **Minimal wastage:** The FDM process builds parts directly by extruding semi-liquid melt onto the model. Thus, only those material needed to build the part and its support are needed, and material wastages are kept to a minimum.
4. **Ease of support removal:** With the use of Break Away Support System (BASS) and Water-Soluble Support System, support structures generated during the FDM building process can be easily broken off or simply washed away.
5. **Ease of material change:** Build materials, supplied in spool form are easy to handle and can be changed readily. This keeps the operation of the machine simple and the maintenance relatively easy.

LIMITATIONS:

1. **Restricted Accuracy:** The diameter of standard filaments (commonly 1.75 mm or 2.85 mm) limits the achievable dimensional accuracy of FDM parts.
2. **Slower Build Speed:** Printing large cross-sectional areas can be time-consuming, as deposition rate is limited by the extrusion speed of the material.
3. **Unpredictable Shrinkage and Warping:** Rapid cooling of extruded material induces internal stresses, which can cause shrinkage, warping, or dimensional distortions in the final part.
4. **Surface Finish Limitations:** FDM-printed parts generally have visible layer lines and rough surfaces, requiring post-processing such as sanding, priming, or coating to achieve smooth finishes.

1.3.3 APPLICATIONS OF FDM

FDM is one of the most widely used additive manufacturing technologies, known for its ability to build durable parts using thermoplastic materials in filament form. Its unique features such as cost-effectiveness, ease of use, ability to process a wide range of engineering-grade polymers, and suitability for functional prototyping make it a versatile AM technique. FDM is especially valued for producing parts with good mechanical strength, material stability, and reasonable accuracy, making it a preferred choice for both prototypes and end-use applications. Following are the potential areas of application:

- **Functional Prototypes** - Fit, form, and functional testing of parts.
- **End-Use Parts** - Low-volume production of customized components and replacement parts.
- **Tooling & Fixtures** - Jigs, fixtures, and assembly tools for manufacturing.
- **Automotive & Aerospace** - Strong, lightweight thermoplastic parts for testing or end-use.
- **Consumer Products** - Low-cost customized parts, housings, enclosures.
- **Educational Use** - Affordable 3D printing for teaching and training.

1.3.4 FDM MATERIALS

FDM filaments are generally thermoplastic polymers supplied in spool form, including materials such as PLA, ABS, PETG, Nylon, TPU, or composites reinforced with carbon fiber, glass fiber. Each filament formulation is designed to provide specific mechanical, thermal, or functional properties, making it suitable for applications ranging from concept models and prototypes to functional and end-use parts.

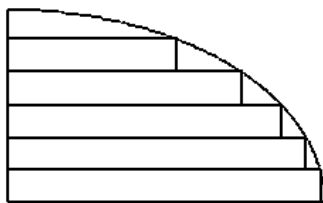
FDM Materials – Composition, Properties, and Applications

Category	Material Type	Key Properties	Typical Applications
Standard Filaments	Polylactic Acid (PLA)	Easy to print, low warping, biodegradable	Concept models, educational models, visual prototypes
	Acrylonitrile Butadiene Styrene (ABS)	Strong, durable, heat-resistant	Functional prototypes, automotive parts, enclosures

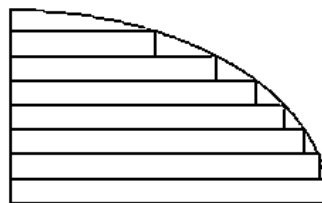
	Polyethylene Terephthalate Glycol (PETG)	Flexible, chemically resistant, moderate strength	Mechanical parts, housings, protective cases
Engineering Filaments	Nylon (Polyamide)	High strength, wear-resistant, flexible	Functional prototypes, gears, bearings, load-bearing parts
	Polycarbonate (PC)	Tough, impact-resistant, heat-stable	Engineering components, automotive, electrical housings
	Polyether ether ketone (PEEK)	High strength, high-temperature resistance, chemical resistant	Aerospace, medical implants, industrial applications
Composite Filaments	Carbon Fiber Reinforced PLA, Nylon, PETG	High stiffness, dimensional stability, lightweight	Structural parts, jigs, fixtures, drones
	Glass Fiber Reinforced PLA, Nylon, PETG	Increased strength, heat resistance	Functional parts requiring durability and stability
Flexible / Specialty Filaments	Thermoplastic polyurethanes (TPU) & Thermoplastic Elastomers (TPE)	Rubber-like, flexible, resilient	Gaskets, seals, wearable devices

1.3.5 PROCESS PARAMETERS

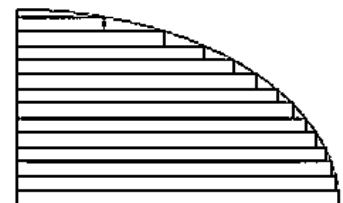
1. Layer thickness:



Thick layer
Rough surface
Less building time



Middle layer thickness
Smooth surface
Moderate Building time less



Very thin layer
Very smooth surface
High building time

Effect of layer thickness on surface finish and part building time

Layer thickness refers to the thickness of the bead extruded from the nozzle tip or the vertical distance (Z-direction) between successive layers. It directly affects build time, surface quality, and resolution, particularly on inclined surfaces. The achievable layer thickness depends on the machine specifications, nozzle diameter, and type of filament material used.

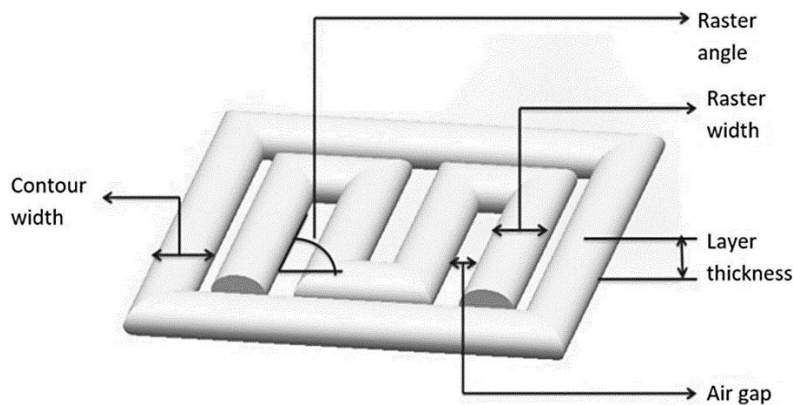
2. Bed Temperature:

Bed temperature influences the adhesion of the part to the build platform. Optimal adhesion is achieved when the bed is slightly above the Glass transition Temperature of the material, as this increases the contact area due to thermal expansion and reduces warping. Excessively high bed temperatures, however, may lead to distortion or sagging of the part.

3. Extruder Temperature:

The extruder temperature controls the viscosity and flow of the molten filament, affecting both interlayer bonding and the surface finish of the part. Higher temperatures generally improve adhesion to the previous layer and produce smoother surfaces, but excessively high temperatures can cause material deformation or stringing. Most FDM machines automatically adjust the extruder temperature based on the material type and maximum allowable limits.

4. Raster and Contour width:

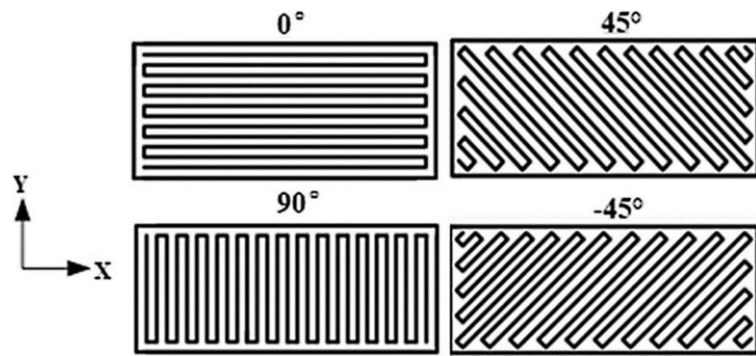


Raster width, Contour width and Raster Angle

- **Raster Width:** This is the width of the extruded bead deposited on the build platform. It is typically 1.2–1.5 times the nozzle diameter and determines the density and strength of the internal infill.
- **Contour Width:** This refers to the width of the outermost perimeter or top layer of the part. A larger contour width is preferred for better surface finish and dimensional accuracy, while thin contours may deform during extrusion. Wider contours are also ideal for hollow or shell structures.

5. Raster Angle:

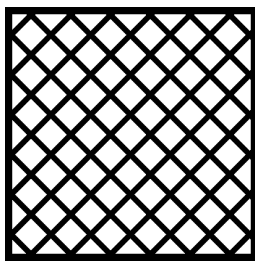
Raster angle is the angle made by the raster tool path with respect to the X-axis of the build plate. The commonly used raster angles are cross ($0^\circ/90^\circ$) and crisscross ($-45^\circ/45^\circ$). The raster angle plays a significant role in the bonding and strength of the parts. In addition, it also influences the surface finish of the built part. The crisscross pattern generally provides better surface finish.



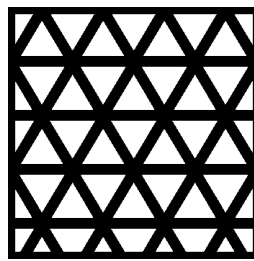
Line Infill pattern with raster angles ($0^\circ/90^\circ$) and ($-45^\circ/45^\circ$)

6. Infill Pattern:

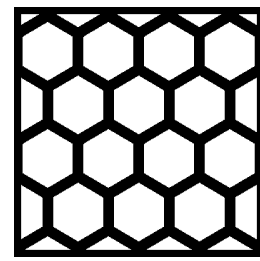
Infill pattern is the structure and shape of the material inside of a part. These can affect a part's strength, weight, print time, and even flexibility. Commonly used infill patterns are:



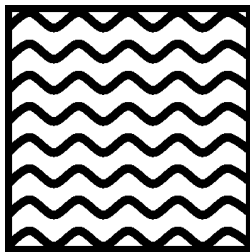
Grid



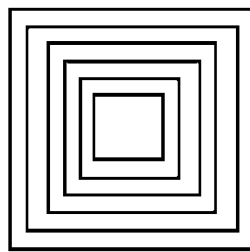
Triangle



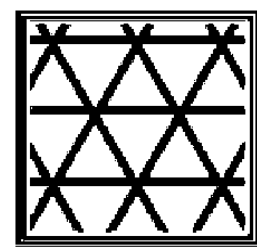
Honeycomb



Gyroid



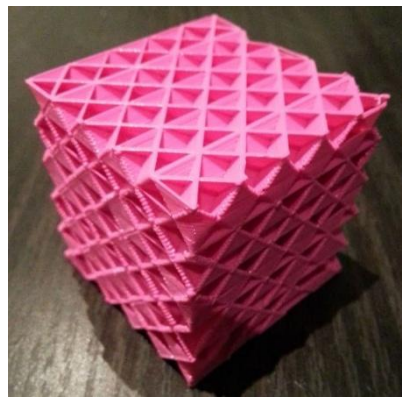
Concentric



Tetrahedral

- **Lines:** The lines infill pattern contains lines printed in one direction (either along the X or Y-axis) every other layer. This infill pattern provides strength in only two dimensions and is good for quick prints. The lines pattern doesn't use too much material and keeps weight pretty light.
- **Grid:** The grid infill pattern is similar in look to lines, but instead of one-directional lines every other layer, it contains two-dimensional lines every layer, with twice as much space in between lines. This pattern provides two-dimensional strength but is still somewhat strong. The grid pattern consumes an average amount of material to complete.
- **Triangles:** The triangles pattern looks like overlapping triangular lines, with lines going in three directions in the XY-plane. This infill pattern provides strength only in two dimensions but still works for prints that need to be strong.

- **Honeycomb:** This pattern produces a honeycomb structure, making for an appealing visual. This infill pattern is good for semi-fast prints that require moderate strength, and it doesn't consume too much material.
- **Gyroid:** The gyroid infill pattern may perhaps look the coolest but it's also arguably the strangest infill pattern. It includes concaving irregular curvatures that eventually cross paths. It's meant to strike an optimal balance between strength, material, and print time.
- **Concentric:** The concentric infill pattern is an internal structure composed of concentric lines that match a part's outline. This pattern is quick to print, good for flexible parts, and consumes significantly less material than most patterns.
- **Tetrahedral:** The tetrahedral infill pattern is very strong 3D structure, lightweight, distributes forces evenly. Used commonly for Aerospace-grade parts, high-strength lightweight applications.



Tetrahedral Infill pattern in 3D